arXiv: 2504.02882 · Tool-Augmented LLMs

# DiaTool-DPO: Multi-Turn Direct Preference Optimization for Tool-Augmented LLMs

Modeling TA-LLM interactions as a 5-state MDP and automatically constructing preference trajectory pairs to improve slot-filling and tool call rejection — without any human annotation.

Direct Preference Optimization · Markov Decision Process · FunctionChat-Bench · Near GPT-4o Performance

PROBLEM · MOTIVATION

# SFT Alone Fails to Teach Slot-Filling and Tool Rejection

THEORETICAL FOUNDATION

# Modeling TA-LLM Dialogue as a 5-State MDP

**$S_1$ — Initial State**
No conversation history. Out-of-scope requests return a rejection and remain here (Type 3 self-loop).

**$S_2$ — Tool Selected — Incomplete Slots**
Tool is identified but required parameters are missing. Slot-filling QA interactions occur here; may repeat N times.

**$S_3$ — Tool Selected — All Slots Complete**
Both tool selection and all required argument values are fully determined. Ready for execution.

**$S_4$ — Wait for Tool Response**
Tool call has been dispatched. System awaits results from the external tool API.

**$S_5$ — Complete**
Tool results received. Final completion message generated and returned to the user.

### STATE TRAJECTORIES

**Type 1**
$S_1 \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$
All args present. Direct path.

**Type 2**
$S_1 \rightarrow (S_2 \times N) \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$
Missing args. N slot-filling turns.

**Type 3**
$S_1 \rightarrow S_1$
Out-of-scope. Self-loop rejection.

States are *latent* — inferred from dialogue context, never directly observed by the model.

QUERY CLASSIFICATION

# Three Query Types Define Distinct State Trajectories

| [illegible] | Condition | [illegible] | [illegible] | [illegible] |
|---|---|---|---|---|
| Type 1 | All required arguments present | [illegible] | [illegible] | [illegible] |
| Type 2 | One or more required arguments missing | [illegible] | [illegible] | [illegible] |
| Type 3 | No matching tool available | [illegible] | [illegible] | [illegible] |

DATASET CONSTRUCTION

# Automatic Paired Trajectory Construction — No Human Labels

| QUERY | ✓ CHOSEN TRAJECTORY | ✗ REJECTED TRAJECTORY | LEARNING LESSON |
| --- | --- | --- | --- |
| Type 1 | $S_1 \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$ | $S_1 \rightarrow S_2 \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$<br>$S_1 \rightarrow (S_2 \times N) \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$ | *Prevent redundant slot-filling questions when all required arguments are already present* |
| Type 2 | $S_1 \rightarrow (S_2 \times N) \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$ | $S_1 \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$<br>$S_1 \rightarrow (S_2 \times M) \rightarrow S_3 \rightarrow S_4 \rightarrow S_5$ $(M \neq N)$ | *Prevent slot hallucination; gather exactly the missing arguments — no more, no less* |
| Type 3 | $S_1 \rightarrow S_1$<br>*(graceful rejection)* | $S_1 \rightarrow S_3 \rightarrow S_4$<br>$S_1 \rightarrow (S_2 \times N) \rightarrow S_3 \rightarrow S_4$ | *Reject out-of-scope requests; never hallucinate a tool call for unavailable functionality* |

FUNCTIONCHAT-BENCH RESULTS

# SFT + DiaTool-DPO Achieves Best Macro Average Across All Models

LLAMA-3-8B · IMPACT OF DIATOOL-DPO

# +43.5% Slot Accuracy and +10.5% Relevance with DiaTool-DPO

SFT-only vs. SFT + DiaTool-DPO on FunctionChat-Bench. LLaMA-3-8B shows the most dramatic improvement.

### Slot Accuracy

1.00
0.50

0.639 SFT-only

0.917 SFT + DPO

↑ +43.5%

### Relevance (Tool Rejection)

1.00
0.50

0.826 SFT-only

0.913 SFT + DPO

↑ +10.5%

KEY TAKEAWAYS

# RL-Based Dialogue Control for Production TA-LLMs

01 **MDP formulation enables systematic preference pair construction.** Defining 5 latent states and 3 query types allows fully automatic generation of chosen/rejected trajectory pairs — no human annotation required.

02 **DPO complements SFT — it does not replace it.** DiaTool-DPO-only performs poorly across all models. Gains come from SFT + DPO together, confirming DPO is a fine-grained corrective signal, not a standalone training recipe.

03 **Slot-filling and tool rejection see the largest gains.** [illegible]